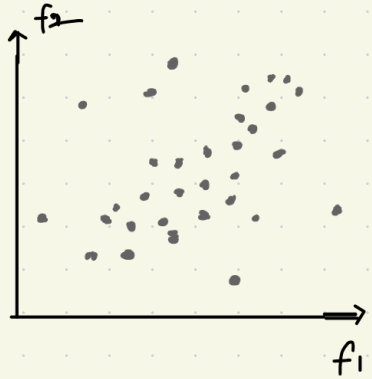
Anomaly Detection



Agenda ⇒
- Anomalies
- iqR / z score
- isolation forest
- LOF

Use case → CARS 24

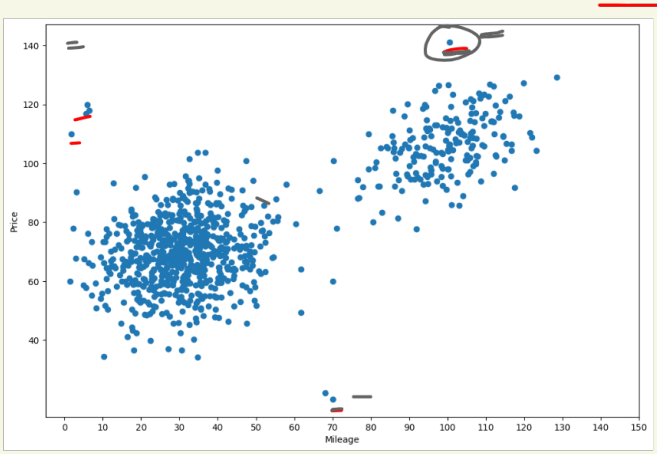
automobile

petrol diesel

EV

mileage , price

Revised data



outliers ?

8 – 10

Anomaly

outliers —

Anomaly → Not normal

Novelty → never happened before

* outlier →

Covid **

* no - of meetings

* existing Techniques →

* IQR Zscore Boxplot DBSCAN → pretty well

* dist^n Based

density **



1. Inter Quartile Range (IQR)

Recall IQR measures variability:

1. **Sort data**: Sorting the data in ascending order
2. **Find the first quartile (Q1)**: Q1 is the median of the lower half of the dataset.
   - If the dataset has an odd number of observations, Q1 is the middle value of the lower half.
   - If the dataset has an even number of observations, Q1 is the average of the two middle values of the lower half.
3. **Find the third quartile (Q3)**: Q3 is the median of the upper half of the dataset.
   - Similarly to Q1, if the dataset has an odd number of observations, Q3 is the middle value of the upper half.
   - If the dataset has an even number of observations, Q3 is the average of the two middle values of the upper half.
4. **Calculate the IQR**: Once you have found Q1 and Q3, the IQR is simply the difference between Q3 and Q1:
   - $IQR = Q3 - Q1$

25th

IQR < Q1 − 1.5 IQR

> Q3 + 1.5 IQR

outliers

2. Z-Score

Remember that **z-score method** can also be used for detecting outliers as:

- It measures how many standard deviations a data point is away from the mean of the dataset.

1. **Calculate the z-score for each data point**: The z-score of a data point (x) is calculated using the formula:

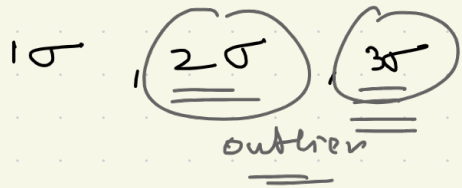
- $z = \frac{x - \mu}{\sigma}$

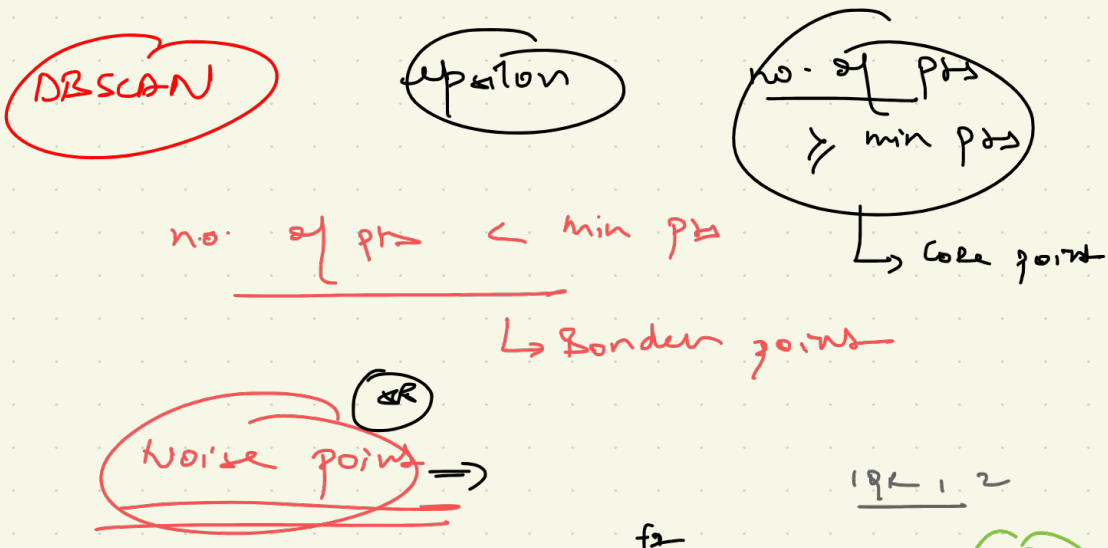
2. **Identify outliers**: Once you have calculated the z-scores for all data points,

- you can identify outliers based on a chosen threshold(τ)
- $|z| > \tau$, the point is an outlier

Note: This method is particularly useful when dealing with normally distributed data.

- A common threshold $\tau = 2, 3$
- If $(|z| > 2)$, the data point is often considered a mild outlier.
- If $(|z| > 3)$, the data point is often considered a significant outlier.





IQR, Z score → similar methods

f1 → BP

Problem → ?

f2 → 2 BP

only work for single / independent features

eg → wt = 10 kg → > 90% of ib

→ outlier

↑

6 ft. 5 inches

BP as a whole = all features

Suppose a distribution with a few data points has a certain mean and standard deviation. If an extreme data point with a positive value is added to the distribution, what will happen to the mean and standard deviation of the distribution parameters?

1 user has participated

A	Mean increases, Standard Deviation decreases	0%
B	Mean decreases, Standard Deviation increases	0%
C	Mean increases, Standard Deviation increases	100%
D	Mean decreases, Standard Deviation decreases	0%

End Quiz Now

10 15 15 10 → SD / 4

= μ ↑

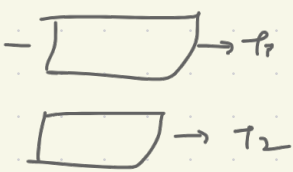
10 15 15 100 = 140 / 4

Isolation forest → Bunch of Trees

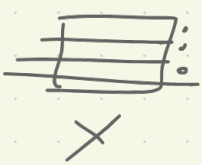
RF → Supervised

Label

Take multiple trees

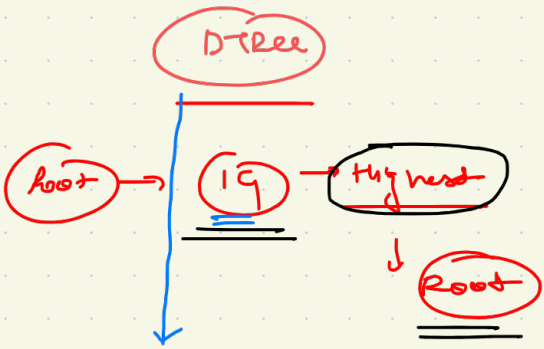


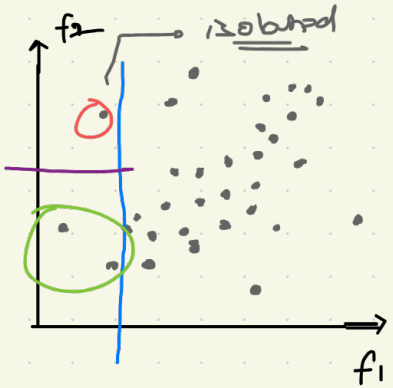
iso forest → isolate → outlier



lot of Trees

isolation forest





axis parallel

Select a feature Randomly

f_1 f_2 → f_1

Threshold for split also decided Randomly

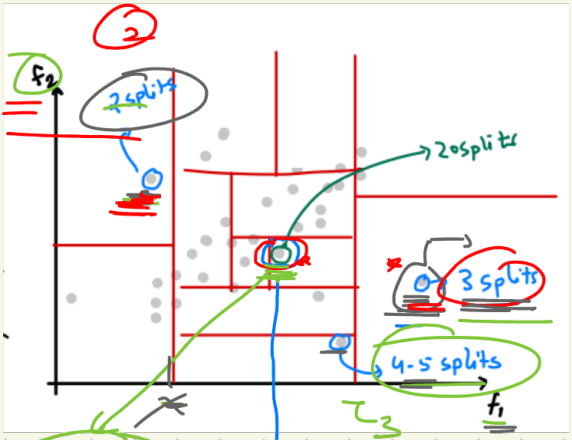
$f_1 < \tau_1$

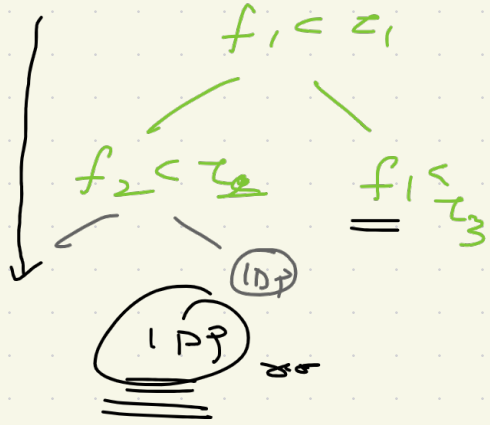
$f_2 < \tau_2$

$f_1 < \tau_1$

$f_2 < \tau_2$







20-30'

8-10

many high no's

splits ↑

node → IDP

OBSERVATION

(1) outlier → lesser splits

(2) inliers → more splits to isolate

splits → depth of DTree







for all DP

(1) Depth of DP →

depth = 1

depth = 2

inliers

depth = 2, 5, 10, 15, 20, 38, 42, 50

outliers

isolate

Random Tree

Random Tree ⟹ single Tree ✗

100s of Trees

100s of Trees → majority

200
300

isolation forest

Bunch of Trees

100s of Trees

Build 100s of Trees

for $\forall x_i \in D$ 100s

depth(x_i) = [4, 8, 6, 5, 4, 2, 7]

depth Req. By 1st Tree

avg depth

avg = 5

outlier

all pt

x_i →

depth (x_i) = [35, 29, 51, 12, 49]

avg = 38.4

inlier

100s

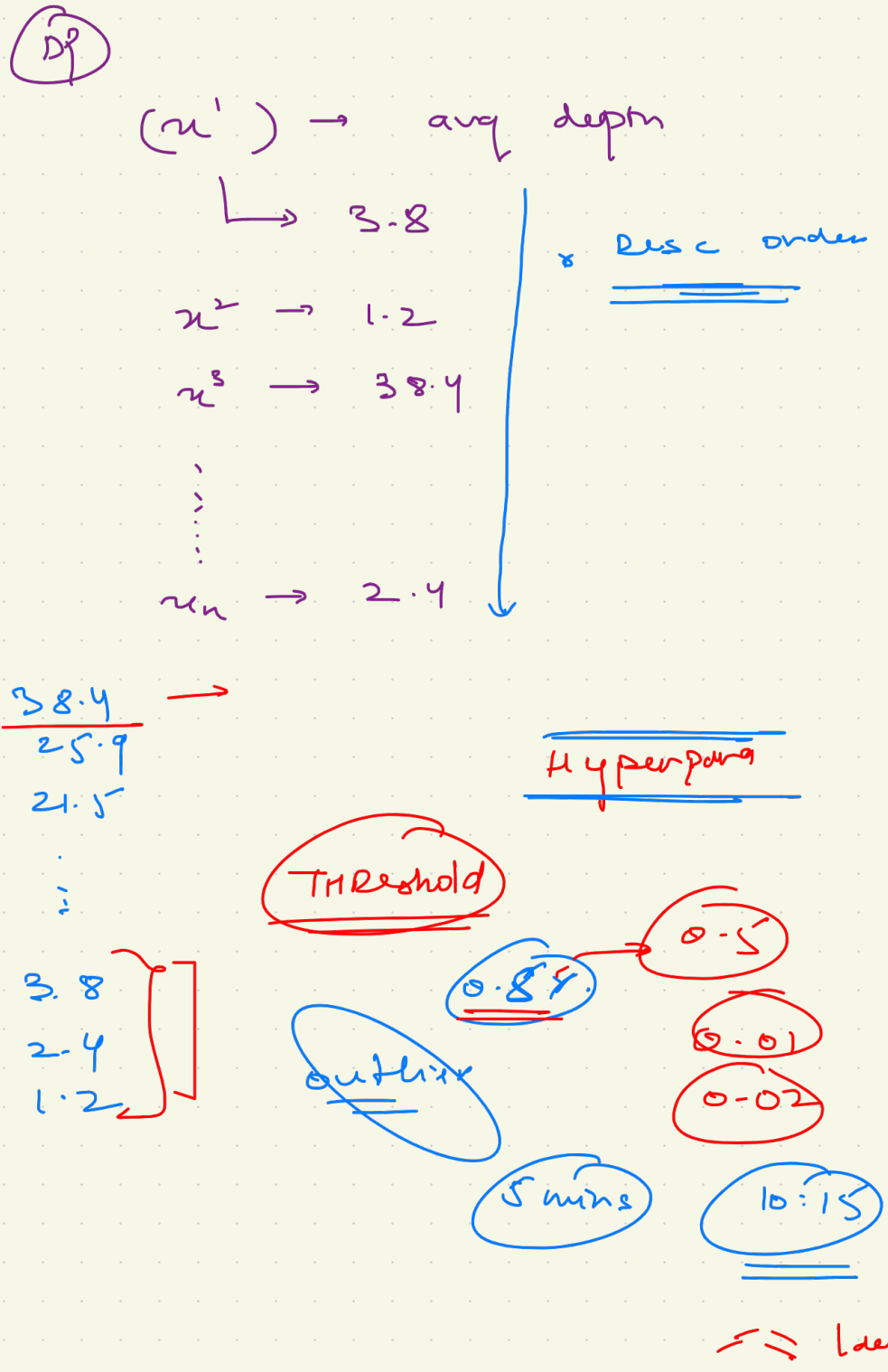
avg depth ↓↓ → easier to isolate

outlier

avg depth ↑↑ → diff. to isolate

inlier DP

isolation forests


Dp
(x') → avg depth
↳ 3.8
x² → 1.2
x³ → 38.4
xn → 2.4
* Desc order
38.4
25.9
21.5
3.8
2.4
1.2
Hyperparm
THReshold
0.85
0.5
0.01
0.02
outlier
5 mins
10:15
== Idepth

In isolation forests, on the basis of what do we select a threshold for a split?

20 users have participated

A	Gini Impurity	0%
B	Entropy	0%
C	Information Gain	5%
✓ D	Random splits	95%

End Quiz Now

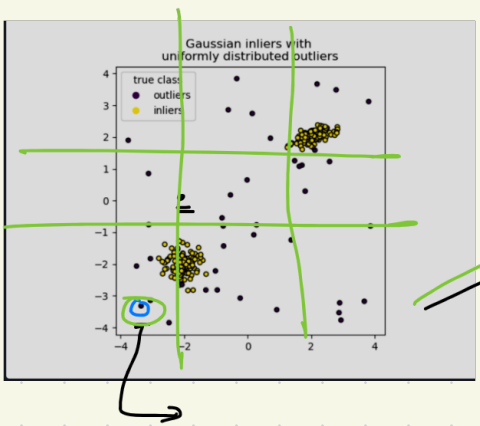
How is the depth of a data point in Isolation Forests related to whether it's an outlier or an inlier?

3 users have participated

A	Outliers tend to have deeper nodes.	0%
✓ B	Inliers tend to have deeper nodes.	33%
C	There is no relation between depth and outlier status.	67%
D	Depth is determined randomly in Isolation Forests.	0%

End Quiz Now

Isolation forest



1 caveat

iso + lof

Passion

Another algo → LOF

(local outlier factor)

KNN + density

Powerful + slow

FAISS, SCANN → indexing

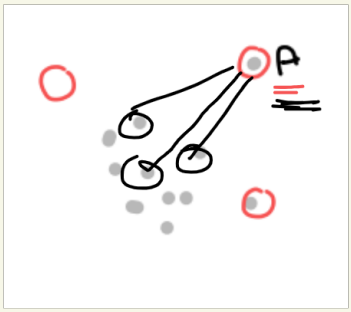
llms

KNN

k=3

Density of A

low

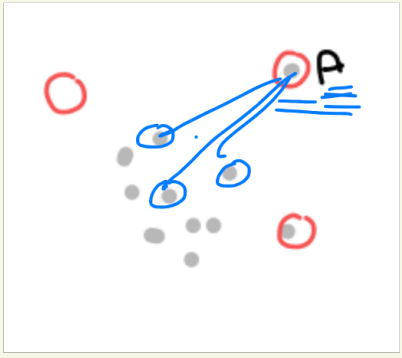


3 outliers

A

intuition → compare pp density with its k neighbors density

3, 4, 5



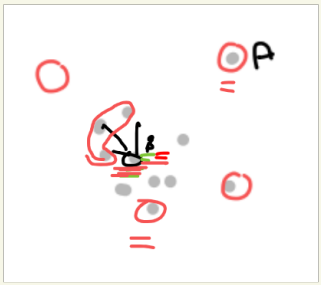
neighbors ↑ density

A ↓ density

Score

A

outlier



$k = 3$

Steps

density of B V/s density of neighbours of B

B → not outlier

inlier

LOF

Steps

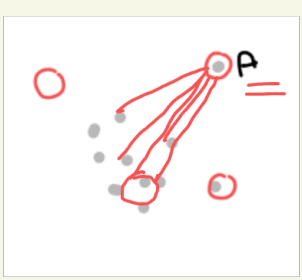
1a

k- k^{th} distance → 5th distance

$k = 5$

5th dist

A



5th neighbour of A

k^{th} neighbour of A

1b

set : $N_k(A) \approx N(A)$ $k=3$

Set of neighbors of point A

k

$\{u^2, u^9, u^{15}\}$ **

neighbor = of A

Step 2

* Reachability Distance

dist (A, B) → euclidean

LOF

** Reach_dist (A, B) = RD(A, B)

$= \max\{\text{euclidean}(A,B), k\text{-dist}(A,B)\}$

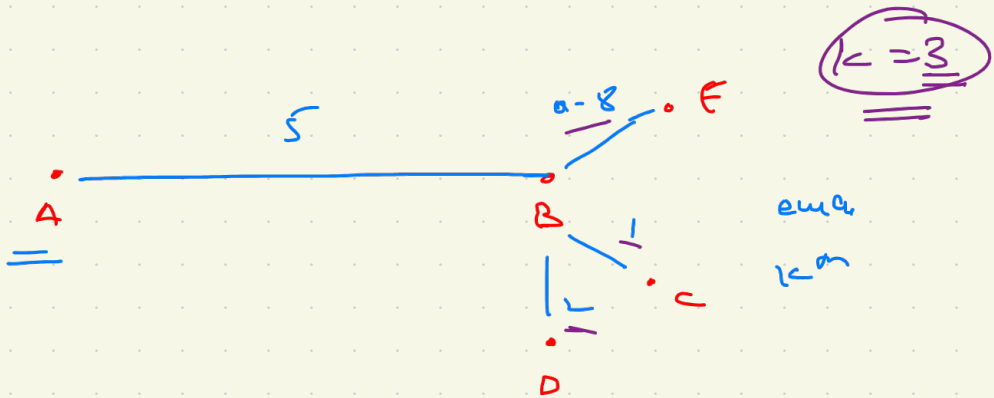
**

Research paper

2 things → euclidean ✓

→ km dist. ✓

RD



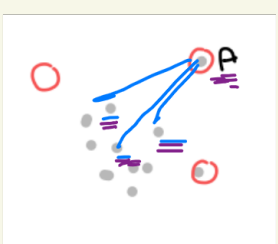
$$RD(A,B) = \max(5, 2) = 5$$

$$RD(C,B) = \max(1, 2) = 2$$

(3) local Reachability Density [LRD]

$k = 3$

$$LRD = \left[\frac{\sum_{B \in N(A)} RD(A,B)}{|N_k(A)|} \right]^{-1}$$



Density of pp = $\frac{RD(A,N_1) + RD(A,N_2) + RD(A,N_3)}{3}$

avg RD

density

(4) Lof score

DP → score

DP (A) = lof (A)

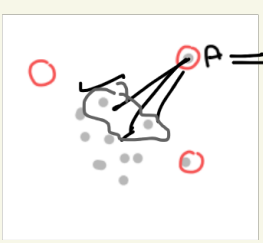
⋮

N → lof (N)

avg density of neighbours of A

$$LOF_k(A) = \frac{\sum_{B \in N_k(A)} lrd_k(B)}{|N_k(A)| \; lrd_k(A)}$$

Density of A

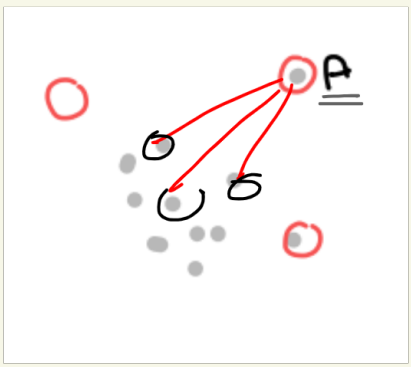


k = 3

density of 1st 2nd 3rd

$$\frac{d_1 + d_2 + d_3}{N_k(A)} = \text{avg density}$$

$$\frac{\text{avg density of neigh (A)}}{\text{density of A}}$$

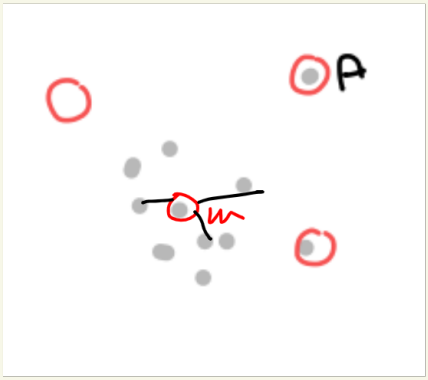


$= \frac{\uparrow\uparrow\uparrow}{\downarrow\downarrow}$ –

lof $\Rightarrow \uparrow\uparrow$

(A) $\uparrow\uparrow$

lof $\uparrow\uparrow \longrightarrow$ outlier



$LOF_k(M) = \frac{\text{avg density}}{\text{density}(m)}$

$= \frac{12}{11} \approx 1$

lof small → inlier

lof Big → outlier

> 1 → outlier

LOF

Eg →

How does LOF determine if a data point is an outlier?

11 users have participated

✓ A	By comparing its density with the density of its neighbors.	73%
B	By using principal components analysis.	0%
C	By calculating its Euclidean distance from the dataset center.	27%
D	By applying a decision tree classifier.	0%

End Quiz Now

Quiz time!

Time Left: 13s

What does a high LOF value indicate about a data point?

14 users have participated

✓ A	It is definitely an outlier.	71%
B	It has the same density as its neighbors.	14%
C	It is definitely an inlier	14%
D	It is a member of a dense cluster.	0%

End Quiz Now